Argued Case for Promotion to Level C (Senior Lecturer) for Bach Le

Overview

In the following, I present my case for promotion to Level C Senior Lecturer in accordance with the University's 'Academic Promotions Guidelines' and with reference to the 'Academic Career Benchmarks & Indicators'. My expertise and scholarship are in the growing area of automated software repair and its broader implications for software engineering theory and practice. I have been developing a strong research track record as evidenced by 35 publications, which received 2020 citations, h-index of 17, i10-index of 22, and the prestigious funding **DECRA 2022 Award worth ~$500K**. Since joining The University of Melbourne in 2019, I have published 21 publications at top and prestigious venues in Software Engineering, and my students achieved multiple awards such as **IEEE Distinguished Paper Award 2023**, **Google PhD Fellowship Award 2023**, and **Best Undergraduate Thesis Award 2019**. My research mission is twofold: (1) to make automated software reasoning/debugging a central part of Software Engineering in ensuring high quality of software, and (2) to innovate in partnership with industry to broaden the research impact via engagements through the Google PhD Fellowship Award and Amazon Applied Scientist Internship program that my students have achieved. My teaching performance at the University is 3.97/5.0 across seven subjects that I have taught, exceeding the requirement of 3.8/5.0 set out by the University. My teaching quality is also reflected via the **University Endeavor Awards in 2020 and 2021** that two teams of master's students under my supervision have won. My annual performance assessment in the last five years has consistently been meeting/exceeding expectations. This document discusses my career in Research and Research Training (60%), Teaching and Learning (30%), and Services and Leadership (10%).

Research and Research Training (60%)

Research Missions and Motivations. Software bugs cost over 1.7 trillion US dollars and affected about 3.7 billion people's lives in 2017. Even a small programming mistake can cause serious adverse effects. It is therefore imperative to uphold the high quality of software, given their pervasive use in our daily lives.

To ensure high quality software, my research revolves around the development of automated techniques and tools for software debugging, guided by actionable insights learned from human and empirical studies. Particularly, my research in human and empirical studies offers deeper understanding of the critical needs in the current software development cycle. Subsequently, actionable insights learned from the studies can be used to realize automated tools/techniques in effective ways to help aid developers in developing more reliable software in a timely manner, further reducing manual costs. Early on, I focused on researching the use of Data Mining, Machine Learning, Software Testing and Verification, Program Analysis and Synthesis to automatically detect and repair software bugs. Fast forward, together with my students, I am also currently exploring the use of automated debugging techniques to guarantee the safety and security of AI models.

Research Impacts and Influences. My most cited first-authored research paper "History Driven Program Repair" [4], has attracted 400 citations, and was adopted by the industry (i.e., the GetAFix bug repair tool by Facebook). GetAFix marks a breakthrough in automated bug fixing, as it is the first ever bug fixing tool successfully deployed in the industry, particularly on Facebook's codebases. The underlying technology of GetAFix relies on a technique that has been directly inspired by "History Driven Program Repair" (as explicitly mentioned in GetAFix's paper). The "History Driven Program Repair" paper remains the most cited paper from the International Conference on Software Analysis, Evolution, and Reengineering (SANER) since 2016.

Automated debugging with a special focus on Automated Program Repair. Ten years ago, automated software bug fixing remained a futuristic idea. This has gradually materialized via several active research projects and breakthroughs in the field that involve large companies such as Meta, Google, and Microsoft, among which my research has made a strong impact and influence. Following on the success of the "History

Driven Program Repair" work and the DECRA'22 award, I continue to push the boundary of automated bug fixing further via publications at top tier conferences and journals such as ICSE, ESEC/FSE, ISSTA, ICSME, TSE, and TOSEM (acceptance rates at around 20-25%).

Together with my students, I built systems that involve mining massive amounts of software artifact data, such as code bug fixes [4, 12, 23, 28, 30], studying human perceptions on software development, such as code correctness and comprehension [13, 16], etc. With insights extracted from those studies, my research offers better understanding of the nature of research problems that we tackle, identifies areas that can be improved and how to achieve them via automated techniques that leverage syntactic and semantics analyses of software programs. Particularly, in addition to the syntactic search such as mining software repositories for syntactic bug patterns used in "History Dirven Program Repair" and the likes, my research also leverages semantics program analysis such as symbolic execution [6, 10, 11], program verification [7, 17, 18], program invariants [28], etc. For example, led by my PhD student, the work *"Invalidator: Automated Patch Correctness Assessment via Semantic and Syntactic Reasoning*" [28] combines syntactic and semantics reasoning of software programs to automatically identify correctness of patches generated by automated bug fixing techniques. Also led by my PhD student, the work "*The Future Can't Help Fix The Past: Assessing Program Repair In The Wild*" [30], which won IEEE Distinguished Paper Award 2023, benchmarks various Automated Program Repair techniques on a new, large scale, high quality and realistic dataset of software bugs collected from the software industry. Other works that are led by me such as *"REFIXAR: Multi-version Reasoning for Automated Repair of Regression Errors."* [20] (during the COVID-19) and "*On Reliability of Patch Correctness Assessment*" [13] leverage automated reasoning on historical versions of software to automatically repair regression bugs in large real-world Java programs and perform human study on how the correctness of machine-generated patches should be assessed. I believe that the synergy between empirical/human studies and the combination of both syntactic and semantics analyses to automated software debugging in general is vital and worthwhile to pursue further.

Exploration of research areas related to automated debugging. I also recently extended my research focus to other related areas such as software security [22, 25, 27, 29], dependable machine learning (e.g., verification of neural networks) [21], and blockchain [16]. My PhD student Thanh-Dat Nguyen has achieved encouraging initial results [21, 23] in these areas that have been accepted to top tier conferences in Software Engineering. We explored the use of Software Engineering techniques such as invariant inference on inferring properties of Graph Neural Networks (GNNs) [21]. In an under-submission work, we showed the effectiveness of such approaches and their numerous applications on ensuring safety and correctness of GNNs such as using inferred properties to effectively detect and defend against attacks on GNNs. Together with my colleague, I also studied challenges and opportunities in smart contract development in blockchain [16], in which I contributed to the design of the study, interviews with senior developers, analyzed the interviews data to disseminate the study results. The work [16] received 542 citations and has been among the top 1% of web of science highly cited papers.

In the next five years, I look forward to closing the gaps between multiple research fields such as automatic bug detection and repair of machine learning libraries and models by leveraging my expertise in automated software repair.

Collaborations. I have been collaborating with world-renowned professors in the field of Software Engineering, both within and outside Australia such as Dr. Chakkrit Tantithamthavorn (Senior Lecturer at Monash University), Prof. Hoa Khanh Dam (University of Wollongong), Prof. Ali Babar (The University of Adelaide), Dr. Xin Xia (Director of Software Engineering and Application Technology Lab, Huawei.), Professor Gary T. Leavens (University of Central Florida), Associate Professor Corina Pasareanu (Carnegie Mellon University and NASA Ames), Associate Professor Claire Legoues (Carnegie Mellon University), Professor Koushik Sen (University of California, Berkeley), Professor David Lo (Singapore Management University), Professor Willem Visser (Senior Principal Applied Scientist at Amazon).

Research Funding. Since my appointment at the University of Melbourne in 2019, I have secured both internal and external fundings of approximately over $500K as sole PI.

My DECRA 2022, titled "*Practical Automated Software Bug Fixing via Syntactic and Semantic Analyses",* brings in an income of ~$500K AUD at The University of Melbourne. Besides, my research also shows good connections with and appreciation by colleagues within the School of Computing and Information Systems at The University of Melbourne, as evidenced by internal fundings awarded by the Computer Science group in CIS such as ECR grant 2022 ($20K, sole PI) and Seed grant 2023 ($8K, with Dr. Christine Rizkallah, Senior Lecturer at CIS). The Seed grant with Dr. Rizkallah resulted in a research proposal that we submitted to Discovery Projects (DP) in 2024 (unsuccessful).

To maintain and foster my international collaboration network, I am also actively involved in helping write international fundings with my close collaborators that attracted ~$2M AUD in total from the Singapore Government, and NSF Medium from the USA (not directly paid to The University of Melbourne). I am also actively involved in engagement with industry funding, as evidenced by a research proposal that was **selected as Finalist funding proposal** (not awarded) by Facebook in 2020: *Agent-based User Interaction Simulation to Find and Fix Integrity and Privacy Issues*, in collaboration with Assistant Professor Tian Yuan (Queens University, Canada).

In the next five years, my goal is to further broaden my collaborations to enhance funding opportunities. For example, Dr. Rizkallah and I plan to resubmit our DP proposal to another suitable venue. Furthermore, together with Dr. Patanamon Thongtanunam (Senior Lecturer at CIS), I plan to submit another research proposal, titled: *"Secured and Robust AI-Augmented Software Quality Assurance",* to Global Science & Technology Diplomacy Fund Strategic Element, 2024 (request for ~$350K). This proposal is based on research papers (under submissions) that Dr. Thongtanunam and I are co-advising our PhD student. I also plan to target industry fundings, such as Google/Amazon Awards in 2025 onwards.

Publications. Since the start of my research career in 2013, I have published in total 35 publications, including 1 book chapter at Automatic Software Verification, 3 x ICSE, 4 x ESEC/FSE, 3 x ASE, 3 x TSE, 1 x EMSE, 2 x ISSTA, 4 x ISSRE, 5 x ICSME 1 x SANER, 1 x ICST, 1 x FoSSaCS, and others at smaller venues. Since joining the University of Melbourne in July 2019, I have published 21 papers, including 2 x ICSE, 2 x ESEC/FSE, 3 x TSE, 1 x ISSTA, 3 x ISSRE, 1 x ICST, 2 x ICSME, and 1 x FoSSaCS, and others at smaller venues. Out of these 21 papers, 12 papers are led (first-authored/co-first authored/second-authored) by my students who are now under my primary supervision (PhD) at the University of Melbourne.

In Software Engineering, prestigious journals include Transactions on Software Engineering (TSE) and Empirical Software Engineering Journal (EMSE). Prestigious conferences with **acceptance rates of around 20%** include Intl. Conf. on Software Engineering (ICSE), Joint European Software Engineering Conf. and Symposium on the Foundations of Software Engineering (ESEC/FSE), Intl. Conf. on Automated Software Engineering (ASE), Intl. Symposium on Software Testing and Analysis (ISSTA). Other good/top venues with **acceptance rates of around 25%** in specialized areas include Intl. Conf. on Software Testing, Verification and Validation (ICST), Intl. Conf. on Software Maintenance and Evolution (ICSME), Intl. Conf. on Software Analysis, Evolution, and Reengineering (SANER), Intl. Symposium on Software Reliability Engineering (ISSRE), Intl. Conference on Foundations of Software Science and Computation Structures (FoSSaCS).

Research Training. I am the primary supervisor of **one postdoc and four PhD student**s (three existing students and one incoming student) and graduated three master's students by research. My postdoc is Dr. Haoye Tian (Google scholar: 409 citations, h-index: 9, i10-index: 8), joined on 1st April 2024. Dr. Tian is

a rising star in the field of Automated Program Repair and is expected to further improve the visibility of my research program on Automated Program Repair at The University of Melbourne.

My PhD students, all of whom under my primary supervision, are Thanh-Dat Nguyen (Full time, Jan 2021 – Dec 2024 expected), Vinay Kabadi (Part time, Jan 2020 – 2026 expected), Cong-Thanh Le (Full time, Feb 2023 – Feb 2026 expected), and Anh Ho (Full time, expect to join in June 2024 on my DECRA'22). My students have achieved outstanding awards under my supervision such as IEEE Distinguished Paper Award 2023 (Vinay Kabadi), Google PhD Fellowship Award 2023 and Amazon Applied Scientist Internship 2024 (Cong-Thanh Le). My most senior PhD student Thanh-Dat Nguyen is on track to submit his PhD thesis by October 2024, having two papers accepted as first author (ICSE'22 and ICSME'22), one paper accepted as second author (ASE'22), two papers under major revisions (TOSEM'24 and ISSTA'24) and another paper currently under submission (TSE'24). My student Cong-Thanh Le has had multiple papers with me (5 papers accepted before PhD, 5 papers accepted during PhD, and two papers currently under review). My student Vinay Kabadi is currently on leave of absence due to family issues and will be returning to study from the end of April 2024. Despite his personal difficulties, Vinay achieved the Distinguished Paper Award at ICSME'23 which motivated him to put his best foot forward in his PhD.

Teaching and Learning (30%)

I have been involved in teaching seven subjects, of which my average teaching performance is 3.97/5.0 overall, which exceeds the requirement of 3.8/5.0 set out by The University of Melbourne.

Approach. For quality and innovation in teaching, I focus on the three strategies: *problem-based learning*, *peer learning* and *curiosity driven learning*, with three key deliveries: *What* – introduce a problem, *Why* – why it is important, and then *How* – how to solve it. To connect these pieces, I always strive to blend theory and practices to help students learn to remember the theory while appreciating their practicality. Sometimes, for more easily explaining the theory, I use analogies in real life as examples to illustrate the ideas. I describe in detail the subjects that I taught, my contributions and innovations below.

SWEN20003 Object-oriented Software Development. Role: Coordinator, undergraduate level, ~350 students. Performance: 4.1/5.0 on average.

Contributions and innovations: New approaches implemented include: (1) several individual meetings with students on their knowledge on the subject and future career, (2) engaging more interactivity via live demonstrations in lectures, e.g., interactive debugging, and via rewarding best student projects with prizes.

COMP90045 Programming Language Implementation. Role: Co-lecturer, graduate level, ~45 students. Performance: 4.4/5.0 on average.

Contributions and innovations: Together with the subject's coordinator, I designed the project assignment which requires students to implement a compiler for a programming language. To refresh the project every year, we introduced new features in the language, e.g., arrays, pointers, etc to enforce new implementation of the compiler. This offers students hands-on implementation skills while learning the theory.

*COMP90082 Masters Software Engineering Project***.** Role: Co-lecturer, graduate level, ~130 students. Performance: 3.9 on average. Especially received 4.35 on useful interaction and engagement opportunities.

Contributions and innovations: Each year we need new industry projects for master's students (10 students per group) to work on. I worked with the coordinator of the subject to ensure that each project is of equal good standard for students to work on. To achieve high quality project and supervision, I worked on hiring project supervisors from the industry, communicating with the supervisors on assessment criteria and monitoring groups' progress via weekly meeting with supervisors.

COMP30026 Models of Computation. Role: Coordinator, undergraduate level, 600 students. This subject is very special in that the coordinator of the subject Professor Harald Sondergaard unexpectedly retired early. As a result, the department asked me for a helping hand, and I thus volunteered to drop my teaching allocation on COMP90045 (45 students) to take over COMP30026 (600 students). It was a tremendous effort for me to help the department deal with this subject. I taught this subject over two semesters as below.

- Semester 1 2021: 3.85 on average. Especially received 4.0 on clear assessments and 4.0 on helpful resources and materials.

Contributions and innovations: We received $10K AUD of funding from the faculty to facilitate blended active learning. We hired tutors to create interactive short dialogues, in which two tutors informally and interactively discuss their views over the content of the lectures each week. We additionally employed the online learning platform Grok for implementing assignments and final exam, which allows students to implement the theory they learned, facilitating **automated assessment** for timely feedback to students.

- Semester 2 2022: 3.6 on average. Note: response rate was very low (60 out of 600)

Contributions and innovations: I regularly improved my knowledge in the subject by reading research papers in the field and adopted advanced concepts to the subject. I adopted changes to the lectures to address some knowledge gap on methods to prove termination and non-termination, rendering students to acknowledge the two sides of the same coin on the long-standing termination problem in computer science.

SWEN90013, SWEN90014, and SWEN90009 over various semesters in 2020 and 2021. Role: Project supervisor that supervises master's students doing projects for industry partners. Performance score is not available as I was the supervisor, not co-lecturer. Under my primary supervision, two (out of four) of my teams have **won University Endeavor Awards in 2020 and 2021** for their innovative master's projects.

Service and Leadership (10%)

Internal services. I served on hiring committee at CIS for hiring faculty members and postdocs to CIS, mentoring peers at CIS on DECRA proposal, and working on international outreach for setting up plans for an exchange program between CIS and Hanoi University of Science and Technology (CIS-HUST), Vietnam. These demonstrate my commitment to helping grow the capacity at CIS in terms of teaching and research. To develop the idea of CIS-HUST student exchange program, I proactively leveraged my long-term collaboration with Prof. Quyet-Thang Huynh, the president of HUST – the largest public university in Vietnam (39,174 full time students, 1,359 faculty members). In March 2024, I invited Prof. Huynh to visit CIS to discuss with CIS's leadership (Prof. Shanika Karunasekera, Deputy Dean, Academic) on possible collaboration plans at school/university level, for which I am appointed by CIS's leadership to coordinate further discussions.

International professional services. I served as PC member/reviewer of prestigious conferences and journals such as ICSE (2020, 2022, 2023, 2025), ASE (2019, 2020), ISSTA (2019, 2020), and journals (various years) such as TSE, TOSEM, TNSM. I also served as chair for Student Research Competition Track at ASE'22, Demonstration Track at IBC'22, and Software Engineering Track at SoICT'23. Being recognized as a leader in the field of Automated Program Repair**,** I was invited to serve on a panel of five world-leading researchers on Automated Bug Fixing to discuss future directions for the research area (organized with ICSE 2023 in Melbourne).

Plans. I am actively looking for opportunities to grow my services to the University and the community. In the next five years, I plan to look for roles in the University such as Program Coordinator of Master of Software Engineering, International Engagements, etc. I also plan to look for roles such as Editorial Board member, and PC chairs of conferences and journals to develop a growing research reputation.